CMSC 475:

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1 Autoencoder

A neural network trained to reproduce its input. Let $F()$ be a composition of two functions: the encoder E and the decoder D:

$$z = E(x), \quad \hat{x} = D(z),$$
$$\text{with the goal } \hat{x} \approx x$$

The objective is to minimize the distance between x and $\hat{x}$, typically through a loss function:

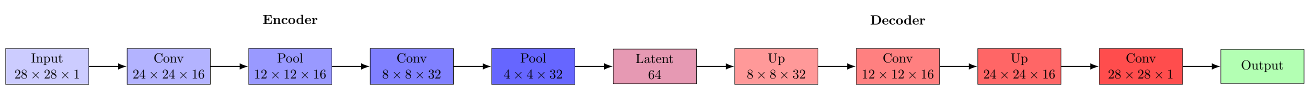
$$\text{Mean Squared Error:}$$
$$\mathcal{L}(x, \hat{x}) = \frac{1}{2} \sum_k (\hat{x}_k - x_k)^2$$

Autoencoders are a form of compression: mapping a high-dimensional vector to a low-dimensional latent vector while retaining information to reconstruct the original input:

$$F : \mathbb{R}^a \to \mathbb{R}^b \to \mathbb{R}^a, \quad a < b \in \mathbb{N}$$

2 Convolutional Autoencoder

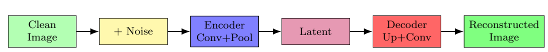
A convolutional autoencoder reduces spatial dimensions via convolutional layers and reconstructs the input using transposed convolutions.



3 Denoising Autoencoder

A denoising autoencoder takes a noisy input $\overline{x}$ and reconstructs the clean image:

$$\overline{x} = x + z, \quad z \sim \mathcal{N}(0, \sigma^2 I), \quad D(E(\overline{x})) \approx x$$



4 Generative Models

There are two primary objectives:

Probabilistic Interpretation

$E(x) \approx P_{\theta_E}(z \mid x)$	(Encoder approximates the posterior of latent variables)
$D(z) \approx P_{\theta_D}(x \mid z)$	(Decoder models the likelihood of data given latents)
$P(x) = \int \int P(x \mid \theta, \phi) P(\theta, \phi) \, d\theta \, d\phi$	(Marginal likelihood of the data)

Reconstruction Objective

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{z \sim E(x)} \left[\|x - D(z)\|^2 \right]$$

So far we have not enforced any "structure" on the latent $\vec{z}$. Without structure, we cannot generate new images from $D()$, because we do not know which latent vectors to sample.

For example, how would you pick $\vec{z}$ to generate an image of an orange-tabby cat?

5 Variational Autoencoder (VAE)

To enable generative modeling, we enforce a prior distribution on the latent space, typically Gaussian:

$$\vec{z} \sim \mathcal{N}(0, I)$$

Properties of Gaussian latent spaces:

- Isotropic: covariance matrix is I
- Dimensions are independent
- Any linear combination of $\vec{z}$ preserves the Gaussian structure

VAEs provide a probabilistic interpretation of autoencoders, allowing us to sample from the model to generate new data. Assume training data $\{x^i\}_{i=1}^N$ and latent variables z.

Data Likelihood

$$p_\theta(x) = \int p_\theta(z)\, p_\theta(x|z)\, dz \approx \frac{p_\theta(x|z)p_\theta(z)}{p_\theta(z|x)}$$

Evidence Lower Bound (ELBO)

$$\log p(x^i) = \mathbb{E}_{z \sim q_\phi(z|x^i)}\Big[\log p_\theta(x^i|z)\Big] - D_{\mathrm{KL}}\big(q_\phi(z|x^i)\,\|\,p_\theta(z)\big)$$

Where:

- $q_\phi(z|x)$: encoder approximating the posterior
- $p_\theta(x|z)$: decoder likelihood
- D_{KL}: KL divergence regularizes the latent space to match the prior

VAE Computation Flow

$$x \to (\mu_{z|x}, \Sigma_{z|x}) \to z \sim \mathcal{N}(\mu_{z|x}, \Sigma_{z|x})$$
$$z \to (\mu_{x|z}, \Sigma_{x|z}) \to \hat{x}$$